

Assignment Sheet

Task 1 – Sign extension

(3 Points)

Explain the concept of sign and zero extension in your own words. What is the difference between these two concepts? When are they used?

Task 2 – NOP's

(2 Points)

What does the NOP-Instruction do? What is its purpose?

Task 3 – Bitmasks

(3+3+3 Points)

All the following operations can be performed by using bitmasks. Explain how.

a) Extract the LSB from a register

b) Invert the content of a register bitwise

- c) Set the MSB to 1

Task 4 – RISC vs. CISC

(2+2+2 Points)

Answer the following questions about RISC and CISC:

- a) What layer is exclusive to CISC?
- b) What could be one reason that pipelining is hardly doable in CISC?
- c) Why was RISC developed when CISC already existed?

This assignment is due on 14.11.2025 at 10 am.